# The psychology of language – E578

The lexical cohort effect

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**Important notion:** Spoken language is fundamentally sequential

# Lexical decision time and nonword point

**Nonword point**: the temporal point from which an item can be unambiguously recognized as nonword.

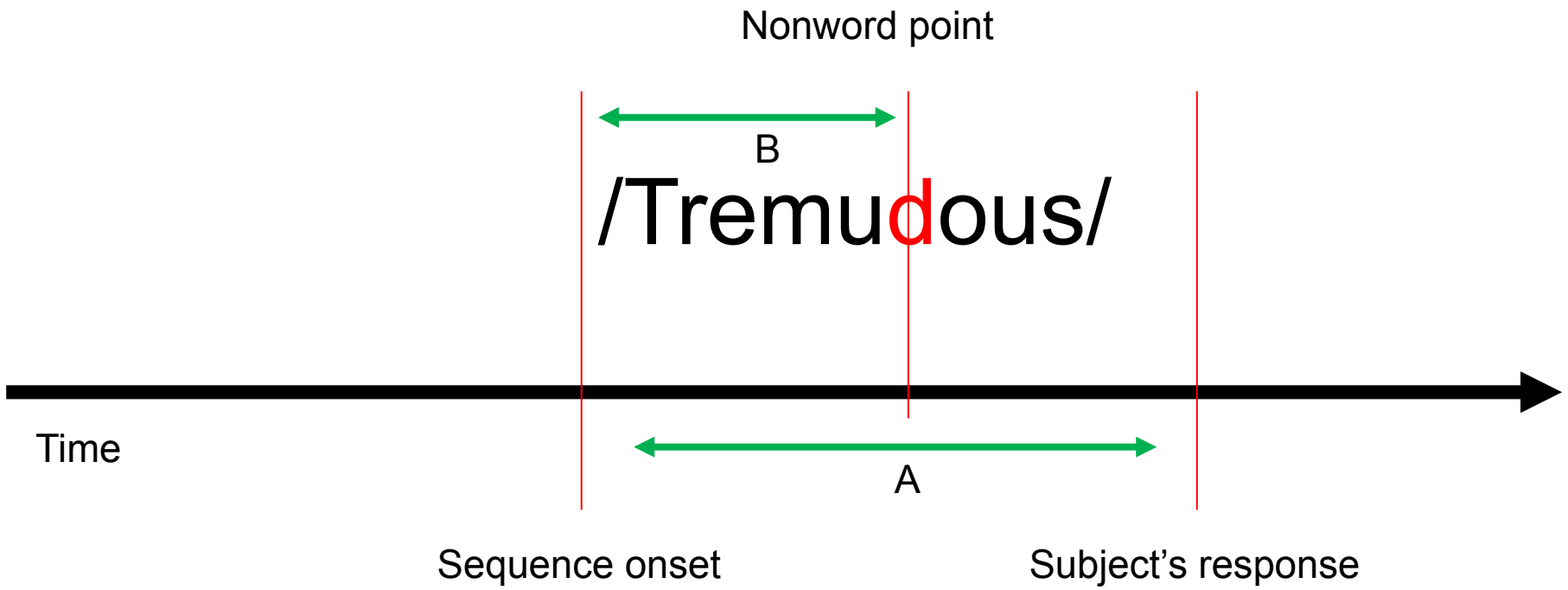
# Lexical decision time and nonword point

| | |
|---|---|
| Treacherous | Tremolo |
| Treachery | Tremor |
| Tread | Tremulous |
| Treadle | Trench |
| Treadmill | Trenchant |
| Treasure | Trend |
| Treasury | Trepidation |
| Treble | Trespass |
| Trek | Tress |
| Trellis | Trestle |
| Tremble | …? |

**Late nonword point:**
Tremulous ➔ Tremudous

**Early nonword point:**
Treacherous ➔ Tveacherous

**Task**: Identify whether the speech input is a word or a nonword (i.e., lexical decision, key press).




A = Dependent variable ➔ Response time
B = Independent variable

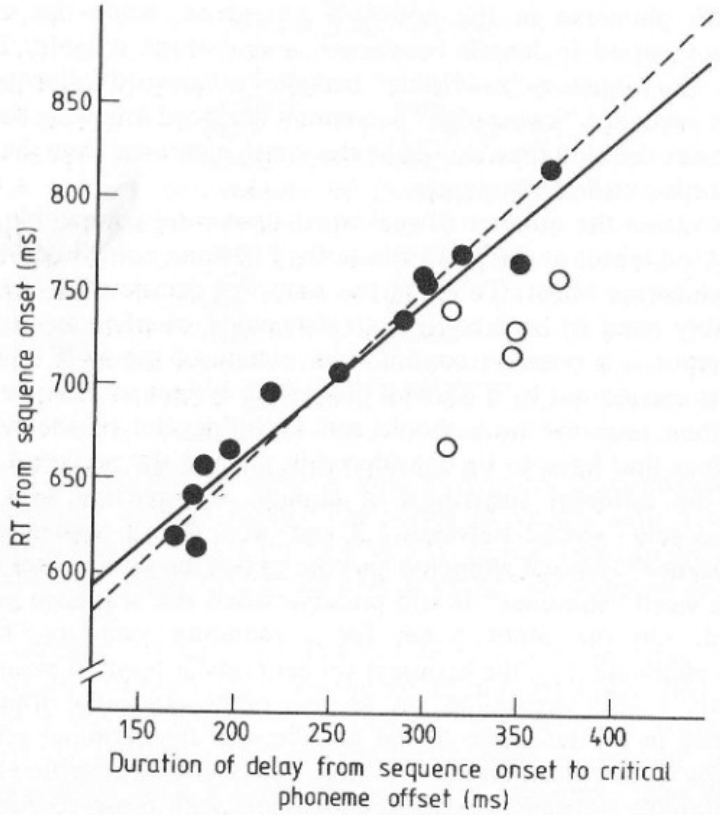



FIG. 8.3. Mean reaction time (from sequence onset) for each syllable-length subgroup, as a function of the delay between the onset of the sequence and the offset of the critical phoneme (the last real-word phoneme). The broken line gives the predicted function (+ 1.0), and the unbroken line gives the observed function (+ 0.90). This is for the filled circles only. See text for discussion of the unfilled circles (low-frequency/small–set-size groups). Each point is the mean of 240 observations.

**Important result:**
Nonword identification slows down as the nonword point increases.

This means that the language recognition system process the speech input progressively in the sequence.

# Lexical decision time and nonword point

We can decompose any word according to a consonant (C) vowel (V) structure:

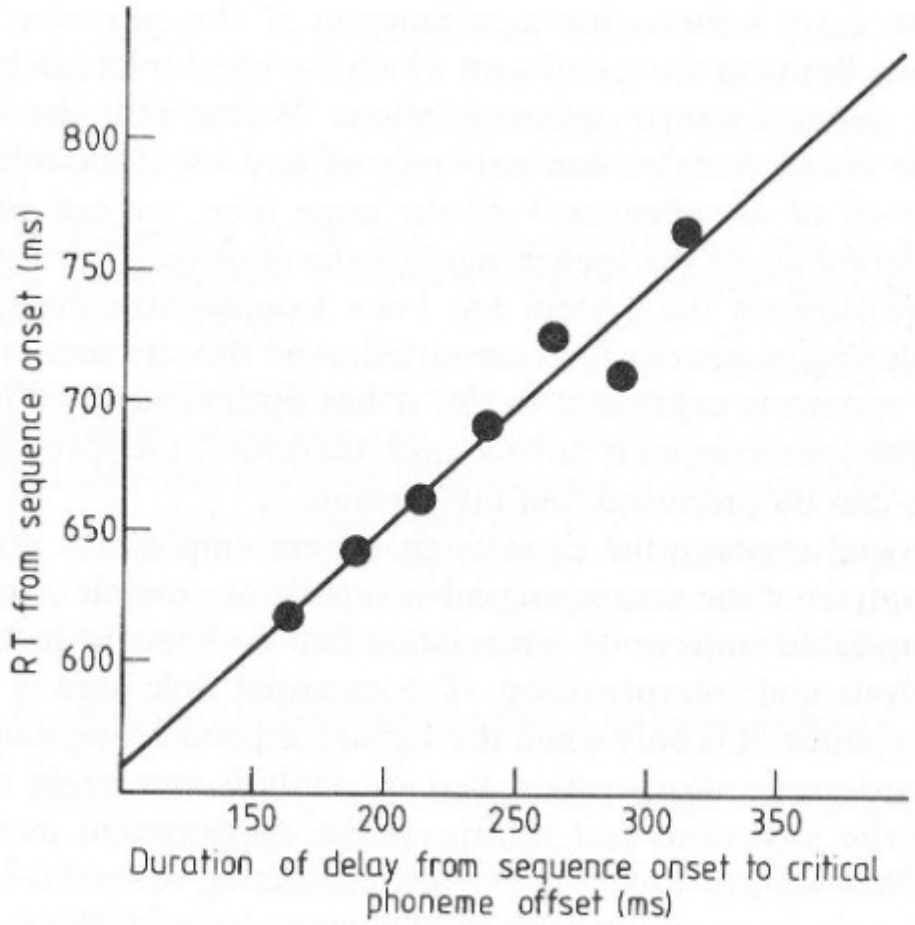
Tre-mu-dous ➔ CCV – CV – CVC

Variations can be done *within* a syllable:

- CCV – CV – CVC
- CCV – CV – CVC
- CCV – CV – CVC

**Question**: Is word recognition performed at the syllable or phonemic level? Do we process **syllables** or **phonemes** in spoken word recognition?

**Prediction**: If word identification is performed at the syllable level, then the critical phoneme within the syllable shouldn't affect decision times in the nonword identification task, because the whole syllable itself should be recognized before reaching a decision.




Response times increase linearly with the moment where the critical phoneme appears.

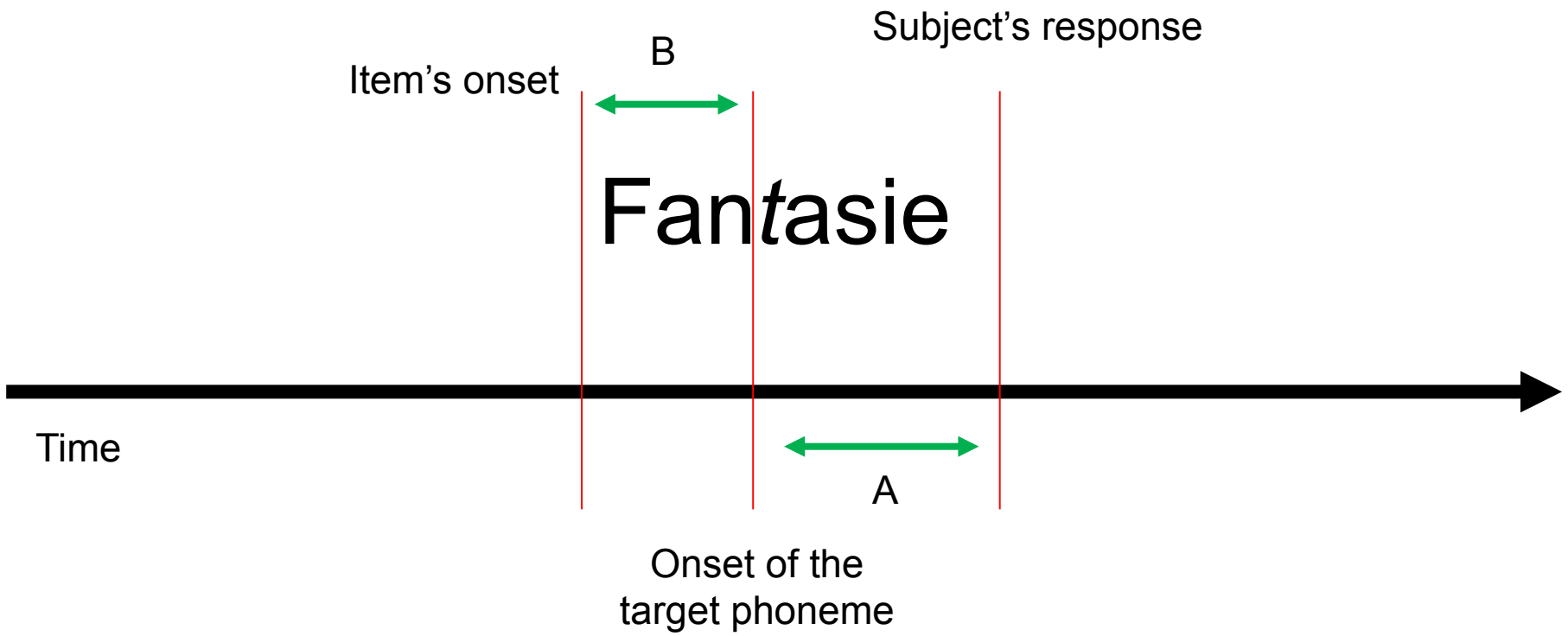
➔ Access to the lexicon is made through a real-time analysis of the speech input signal.

➔ It is performed **online**

# Phoneme monitoring

| Real words | Pseudowords |
|---|---|
| *T*rein | *T*ruig |
| Flui*t* | Zwoo*t* |
| *T*afel | *T*eurpen |
| Ke*t*el | Wie*t*el |
| Bewus*t* | Korbes*t* |
| *T*ekening | *T*ogening |
| Fan*t*asie | Pol*t*elisch |
| Momen*t*eel | Inden*t*risch |
| Kandidaa*t* | Barakaa*t* |

**Task**: to respond as soon as the phoneme /t/ appears. Participants performed the task for words and nonwords in **different blocks**.




A = Dependent variable ➔ Response time
B = Independent variable

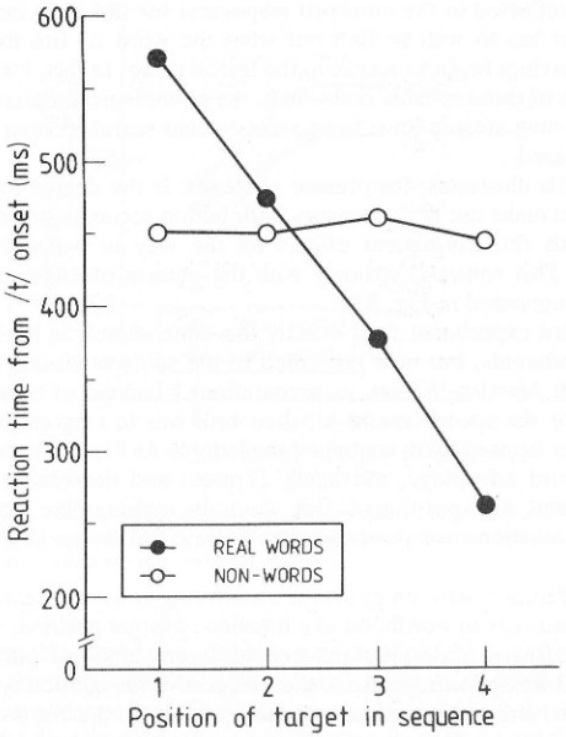



FIG. 8.1. Mean phoneme-monitoring reaction time for real words and nonwords, plotted according to target position and collapsed across syllable lengths. Position 1 corresponds to sequence-initial targets, and Positions 2–4, respectively, to targets at the end of the first, second, and third syllables of the sequences. Note that the number of observations varies across positions. All items contribute to Positions 1 and 2, but only two- and three-syllable sequences contribute to Position 3, while Position 4 applies only to the three-syllable sequences.

**A few observations:**

- Response times are slower for words (compared to nonwords) when the target appears early
- As the target appears late in the processing stage, response times dramatically drop
- Response times as a function of the target position remains fairly unchanged for nonwords

**Implications:**

- People use different routes to process words and nonwords
- As more information is carried by the input signal, the detection of phonological information is made faster for words ➔ people make predictions
- Since no lexical information is available for nonwords, people can’t make predictions for these items

Replication with **visual** presentation, i.e. the items appear in a written format.

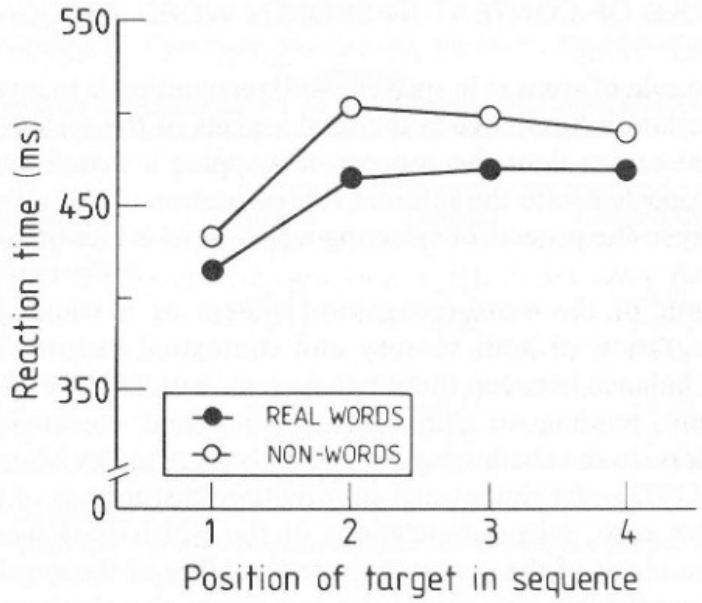



FIG. 8.2. Mean letter-monitoring reaction time for real words and nonwords, plotted according to target position and collapsed across sequence lengths. For further details see caption to Fig. 8.1.

Reading is made in **parallel**, contrary to spoken language comprehension which is inherently sequential.

We will return to this idea later

# The cohort effect

**The size of a cohort** refers to the number of words sharing the initial phonemes in the speech input.

| | | |
|---|---|---|
| Treacherous | Trek | Trend |
| Treachery | Trellis | Trepidation |
| Tread | Tremble | Trespass |
| Treadle | Tremolo | Tress |
| Treadmill | Tremor | Trestle |
| Treasure | Tremulous | … |
| Treasury | Trench | |
| Treble | Trenchant | |

# The cohort effect

**The size of a cohort** refers to the number of words sharing the initial phonemes in the speech input.

We can contrast items associated with small and large cohorts, and check the impact of this variable on response times.

**Task**: Lexical decision.

**Manipulation**: Large vs. small cohorts

Lexicality effect: faster response times for words vs. nonwords

Cohort effect: faster response times for items drawn from small vs. large cohorts

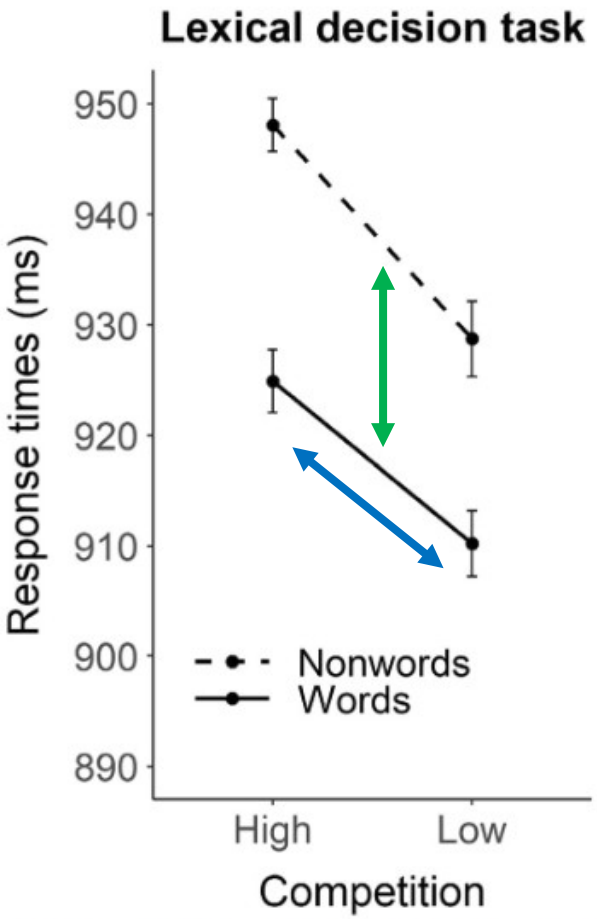



**Fig. 2** Experiment 2 – Mean response times averaged across participants (y-axis) for high- and low-cohort stimuli (x-axis) separately for words (solid line) and nonwords (dashed line). Error bars represent standard errors, after correction for between-subject variability (Cousineau, 2005)

# Neuroimaging evidence

1. Putting people in an MRI
2. Making them perform lexical decisions on a bunch of items

# Neuroimaging evidence

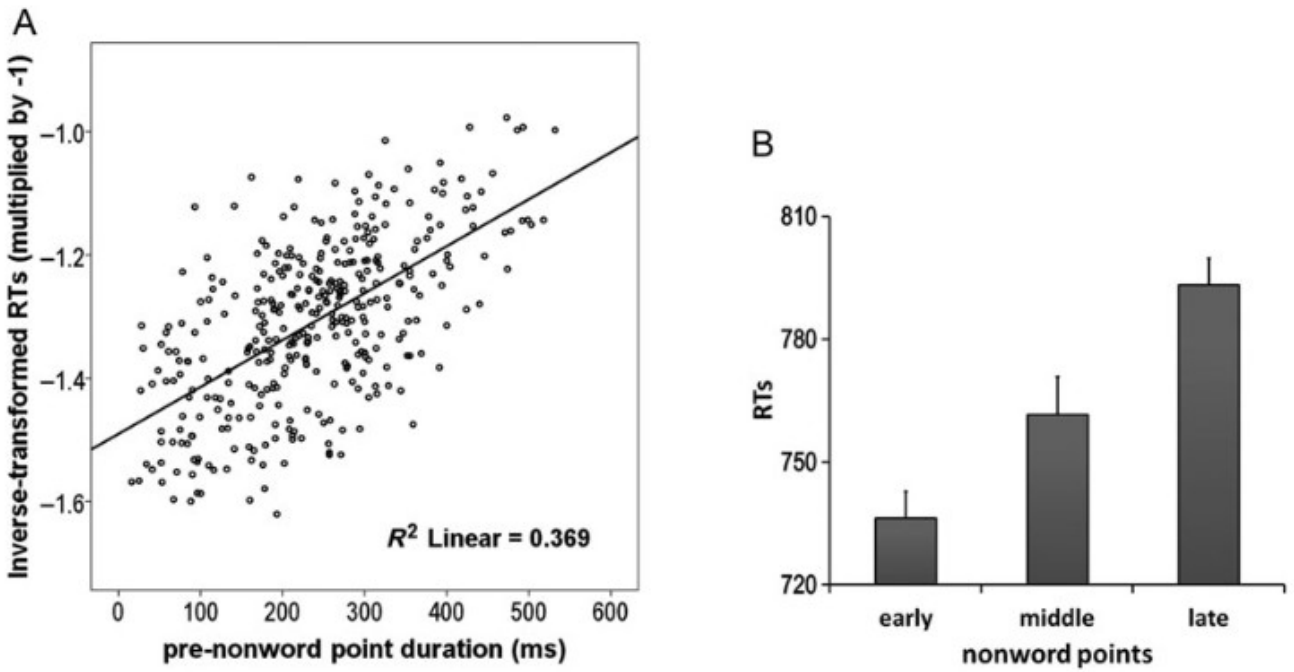



**Figure 1.** Results of behavioral study. (*A*) Item-wise pre-nonword point duration plotted against RT. (*B*) Mean lexical decision RTs at early, middle, and late nonword points.

# Neuroimaging evidence

Parametrical modulation of the activity in phonological – lexical regions as a function of nonword point.

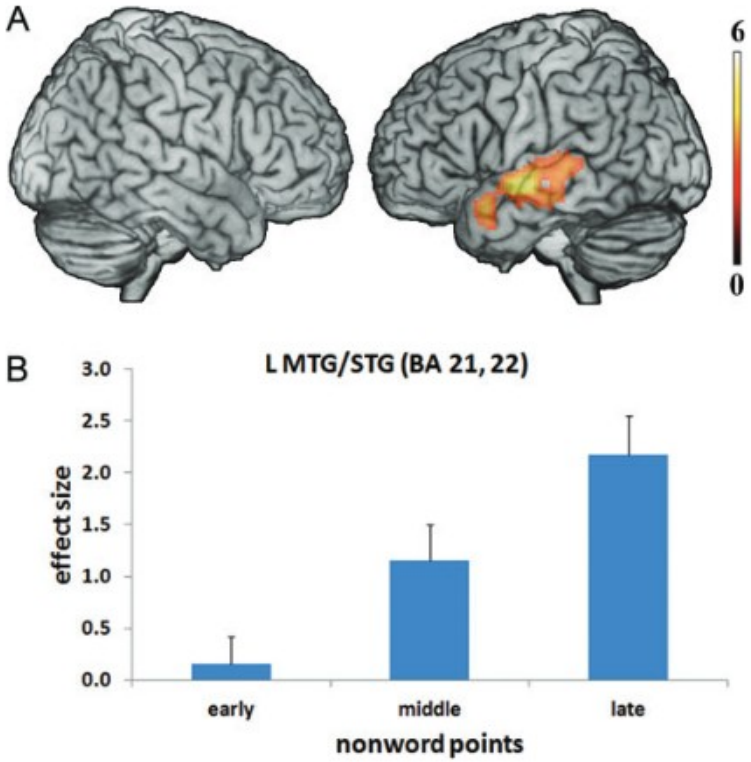



**Figure 4.** (*A*) Increasing activation for later nonword points at a threshold of $P < 0.005$, voxel-level uncorrected, and $P < 0.05$, cluster-level corrected. (*B*) The mean activation value of the significant cluster in (*A*) for each nonword point condition.

# Neuroimaging evidence

Correlation between activity in prefrontal regions and cohort competition / selection.

Those regions are typically involved in controlled processes.

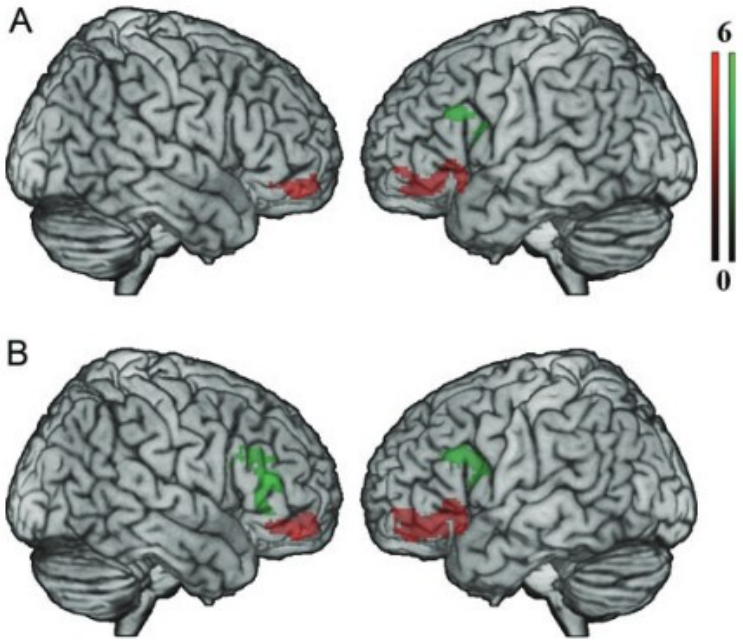



**Figure 5.** (*A*) Significant activation for increasing cohort competition (red) and increasing cohort selection (green) at a threshold of $P < 0.005$, voxel-level uncorrected, and $P < 0.05$, cluster-level corrected. (*B*) The cohort competition and selection effects rendered at a lower threshold of $P < 0.01$, voxel-level uncorrected, and $P < 0.06$, cluster-level corrected.

# Top-down influence of spoken word recognition

Word recognition is not solely driven by the bottom-up input. It also depends on the top-down, contextual information.

**Word monitoring task**: people need to press a button as soon as they hear a target word. The target word is given to them before they hear the sentence.

# Top-down influence of spoken word recognition

Word monitoring task – Late word position

**Target word**: cigarette

**Normal context**: He was stressed, so he decided to light up a *cigarette* before leaving

**Syntactic context**: Several bunch a moon, the cars fly under the *cigarette* that makes ketchup

**Random order**: Leaving he stressed light was, so he to *cigarette* up before decided a

# Top-down influence of spoken word recognition

Word monitoring task – Early word position

**Target word**: cigarette

**Normal context**: The good old *cigarette* is still bad for your lungs

**Syntactic context**: Around the praying cigarette, caucasians glasses navigate

**Random order**: bad Lungs *cigarette* the old is still good for your

# Top-down influence of spoken word recognition

Manipulations:

**The type of context**: normal – syntactic – random

**Word position in a list** – parametric modulation: early – middle – late

**Word duration** – the longer a word, the longer it takes to recognize it

Three variables – Relatively complex data to process

# Top-down influence of spoken word recognition

TABLE 8.2
Contributions of Contextual and Sensory Inputs to Word-Monitoring Performance:
Regressions of Monitoring RT on Word Position and Word Duration[a]

| | *Prose contexts* | | |
|---|---|---|---|
| *Regressions* | *Normal* | *Syntactic* | *Random Order* |
| On word-position: | | | |
| $r =$ | −0.89 | −0.72 | +0.01 |
| $b =$ | −9.44 | −6.46 | +0.06 |
| On word-duration: | | | |
| $r =$ | +0.48 | +0.74 | +0.90 |
| $b =$ | +0.19 | +0.32 | +0.52 |

$r$ = correlation coefficient
$b$ = slope (for the word-position analysis, the slope is in msec per word position; for the word-duration analysis it is in msec per msec increase in word length).

[a]The figures given here are the averaged effects from Experiments One and Two in Marslen-Wilson and Tyler (1980).

Response times decreases with word position… and this is especially the case as the context is relevant in the sentence.

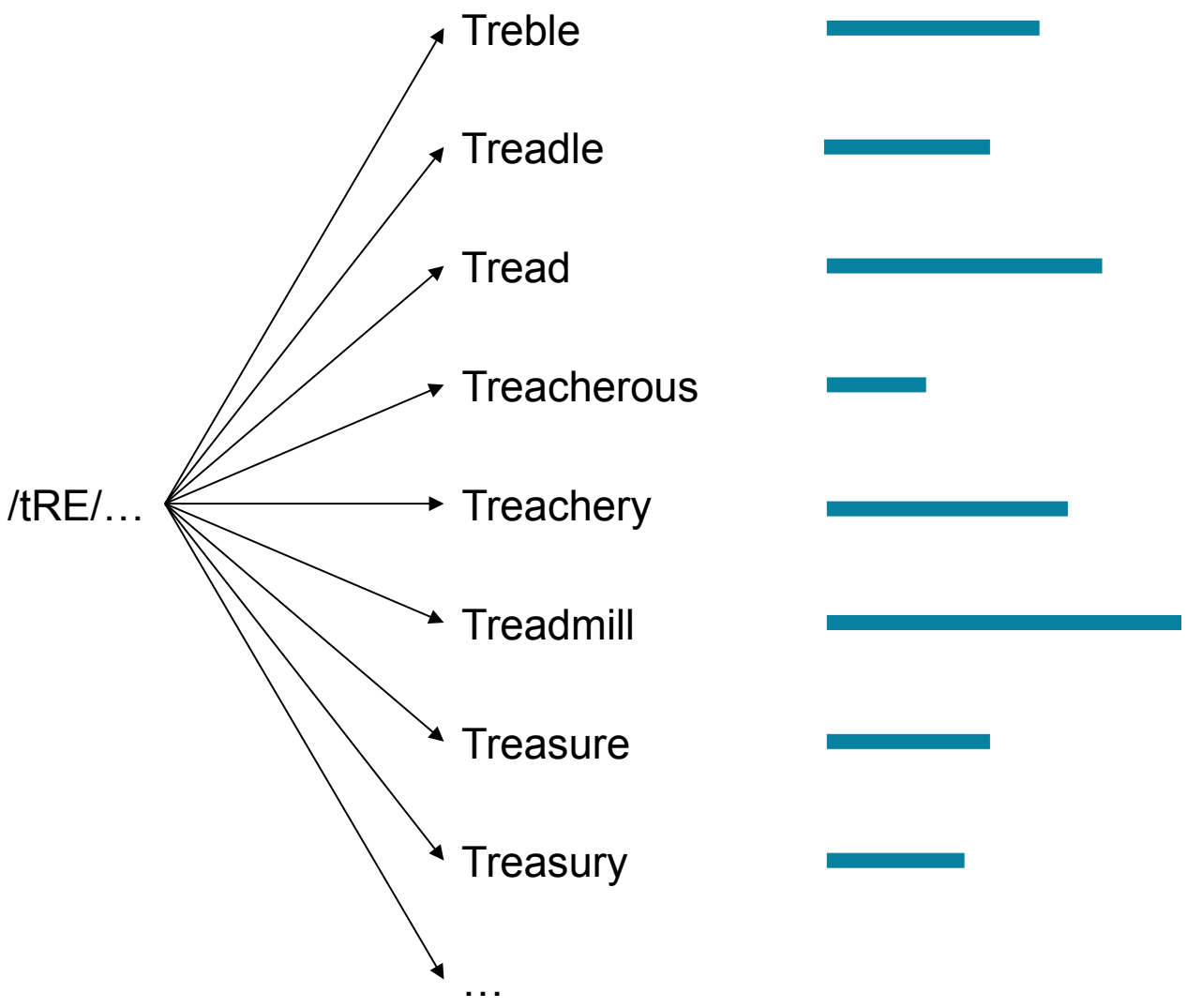
When context is not relevant, word position doesn't matter.

Response times increase with word durations, and this is mostly the case when the context is not relevant.
➔ Effect of the speech input signal are more prelevant when there is a lack of contextually-relevant information.

# Top-down influence of spoken word recognition

Hence, top-down information modulates to what extent spoken recognition is driven by the bottom-up speech input.

**In other word, lower levels of language processing are modulated by people's expectations/precitions!**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Thank you for you attention